Insecure by Design: A Hands-On Learning Platform for Web Security Training

Progress Report 3

CSIS 4495-003 | Applied Research Project

Winter 2026

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Work Logs (Mar 3 – Mar 9)

Date	Number of Hours	Description of Work
Mar 3	1	Met with the instructor for a project check-in. She provided feedback and suggested adding a user profile progress tracker to the platform, allowing users to see which challenges they have completed. Discussed how this could integrate with the existing ChallengeProgress entity.
Mar 4	1.5	Met with Allwyn to discuss new feature ideas following the instructor meeting. Together we conceptualized a "Why It Worked" Explainer feature - a post-challenge explanation section that breaks down why a given payload succeeded, reinforcing the learning outcome for each vulnerability.
Mar 6	1	Emailed the instructor with a brief description and mockup of the "Why It Worked" Explainer concept. Received approval to proceed with the feature as part of the platform.
Mar 6	1.5	Friday team meeting with Allwyn. Reviewed homepage progress, discussed the SQLi level-hiding fix approach, and planned next steps for both the user profile progress tracker and the "Why It Worked" Explainer features.
Mar 7	1.5	Saturday team meeting with Allwyn. Continued discussion on the user profile progress tracker design and the "Why It Worked" Explainer structure. Agreed on implementation approach and divided upcoming tasks.
Mar 8	2	Worked on the homepage redesign. Updated the Thymeleaf index.html template with a hero section, Bootstrap challenge cards, and a features section. Also resumed work on the SQLi level-hiding fix - investigated using server-side th:if conditions to fully hide Level 2 and Level 3 from the rendered HTML until prerequisites are met.
Mar 9	2	Continued homepage refinements and completed the SQLi level-hiding implementation. Updated SqlInjectionController to pass per-level completion flags to the Thymeleaf model and updated sqli.html accordingly. Tested all three challenge levels end-to-end. Finalized and checked in all files for Progress Report 3.

Summary Description

This reporting period began with two key planning sessions. On March 3, I met with the instructor for a project check-in, during which she suggested adding a user profile progress tracker so users can see which challenges they have completed. On March 4, I met with Allwyn and we jointly conceptualized a new feature called the “Why It Worked” Explainer, a post-challenge section that explains why a given payload succeeded, reinforcing the learning outcome. On March 6, I emailed the instructor with a description and mockup of the Explainer concept and received approval to proceed.

Our regular Friday and Saturday team meetings were held on March 7 and 8. We reviewed homepage progress, aligned on the SQLi level-hiding fix approach, and planned the implementation of both the user profile progress tracker and the “Why It Worked” Explainer for upcoming reporting periods.

On Sunday and Monday I focused on two implementation tasks. For the homepage, I redesigned the Thymeleaf index.html template with a hero section, Bootstrap challenge cards, and a features section. For the SQLi challenges, I completed the level-hiding logic that was left unfinished in PR2 — replacing the client-side disabling approach with server-side th:if conditions so that Level 2 and Level 3 are fully absent from the rendered HTML until their prerequisites are met. End-to-end testing confirmed that all three levels work correctly and that progress persists through the ChallengeProgress entity.

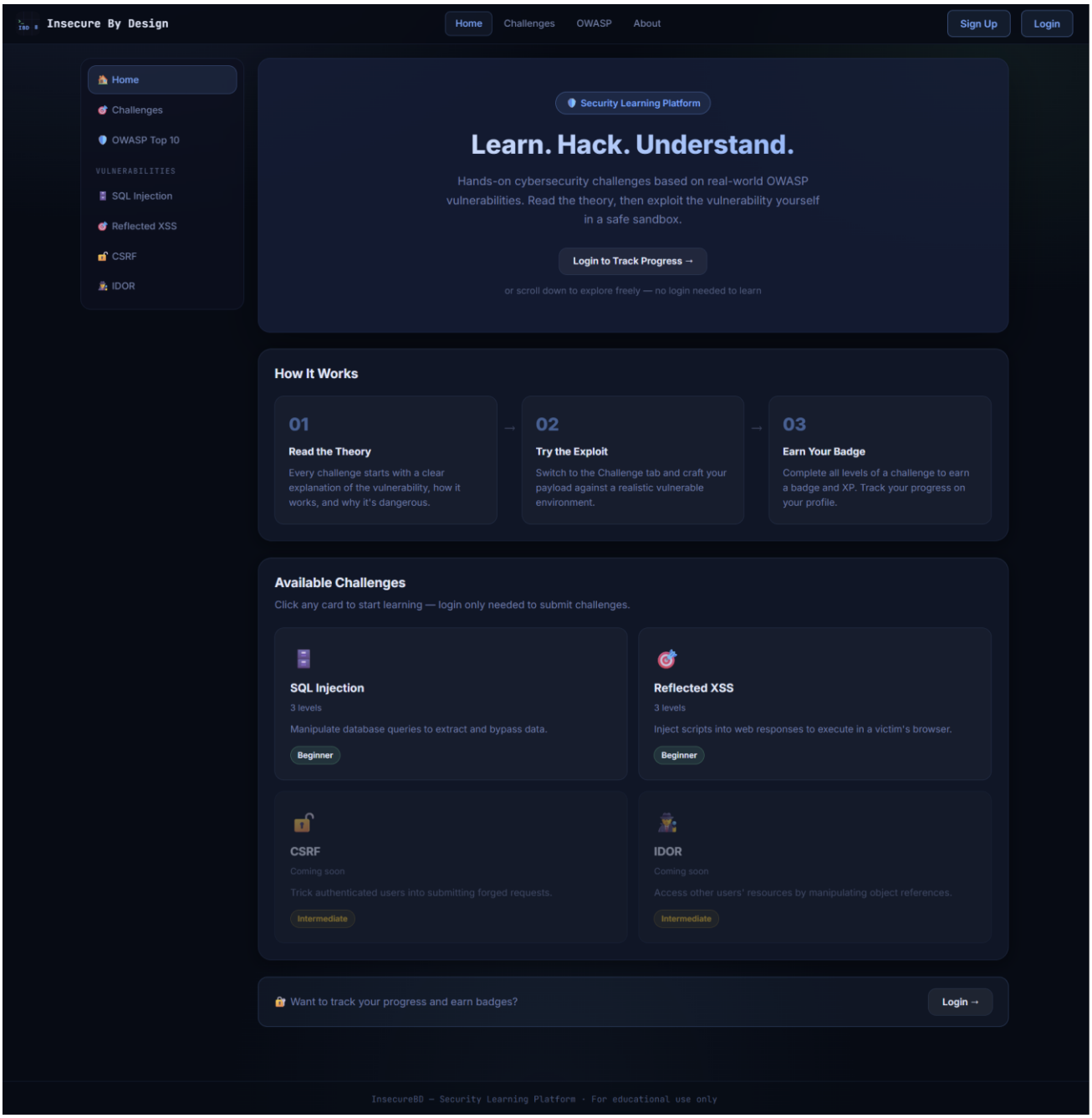
Repo Check-In of Implementation Completed

Since the previous progress report, I have checked the following files and updates into the GitHub repository under the Implementation folder:

- Updated index.html (homepage Thymeleaf template) with redesigned hero section, challenge cards, and features section.
- Custom CSS additions within the homepage template for hero styling and layout adjustments.

- Updated SqlInjectionController.java to pass per-level completion flags to the Thymeleaf model for server-side level visibility control.
- Updated sqli.html with th:if conditions to fully hide Level 2 and Level 3 sections until prerequisites are completed.
- Progress Report 3 (this document) checked into the ReportsAndDocuments folder.

All files were committed individually under my own GitHub account, in compliance with the requirement for individual contributions.



AI Use Section

Table of AI Tools and Specific Use

AI Tool Name	Version, Account Type	Specific feature for which the AI tool was used	Value Addition - What value did you add over and above what AI did for you?
ChatGPT	GPT-4o, free	Asked for suggestions on Bootstrap layout patterns for the homepage hero section.	Evaluated the suggestions, selected only what fit our Thymeleaf structure, and manually integrated and adjusted the layout.
Claude (Anthropic)	Claude Sonnet 4.6, free	Used to assist with coding the homepage redesign (index.html Thymeleaf template) and the SQLi level-hiding logic (SqlInjectionController.java and sqli.html th:if conditions).	Reviewed and tested all generated code against the existing Spring Boot project structure. Identified integration issues, adjusted the controller model attributes to match our entity design, and verified end-to-end behaviour through manual testing.

Appendix

Prompt 1 (ChatGPT, GPT-4o free):

"I am building a Spring Boot web application with Thymeleaf templates. What are some Bootstrap 5 layout patterns I can use for a hero section on a homepage that includes a title, subtitle, and two call-to-action buttons?"

Prompt 2 (Claude, Sonnet 4.6 free):

"I have a Spring Boot controller called SqllnjectionController that passes challenge level data to a Thymeleaf template (sqli.html). Right now, the locked levels are disabled client-side but still visible in the HTML source. How can I pass per-level completion flags through the model and use th:if in Thymeleaf to fully hide Level 2 and Level 3 until the previous level is completed?"